SIMATS ENGINEERING

ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Dave's Taxonomy: Imitation (Level 1) **Total Marks: 25**

Question Count: 5

Code of Conduct

I VIKRAM P /192511321 (Name / Reg. No.) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Vikram P

Assessment Tasks

Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Discuss the role of **data centers** in delivering cloud services..
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

Total Marks:

CO1-AT-1

1. Define cloud computing and explain four characteristic

Ans:

Cloud computing means using computing Services like storage Service and Software through the internet intead of maintining them on our own computers.

1. on-demand Self-Services:
users can access Service wherever they need them.
eg: Creating a virtual machine Aws

2. Broad network access:
Cloud Services can be accessed through mobiles, laptops and tablets using the internet.
eg: Google Drive.

3. Resource Pooling:
Cloud Service Provide share computing resources among many users.
eg: multiple Companies using a cloud data center.

4. Rapid elasticity:
Resource can be increased or decreased to demand.
eg:
An online shopping website adding Servers During festival Sales

Conclusion:

Cloud computing Save cost and Providers flexible and easy access to computing resaurce.

2. Explain the evolution of cloud computing:

Ans:

Cloud Computing developed through Different stages

- hardware evolution:
  Early computers were large and expensive.
- Mainframe Computing:
  many users shared one Powerful Computer.
- Personal Computers:
  Computers became Smallers and affordable.
- client-Server Computing:
  client received computer across from Servers
- Internet evolution:
  The internet connected Computer across the world
- Virtualization:
  one Physical Server Could run multiple virtual machines
- web Software:
  Applications became accessible through web browsers
- modern cloud computing:
  Computing resources and Software became available through the internet

Conclusion:

hardware development, the internet and virtualizing helped in the Development of modern cloud computing

3. Explain Server virtualization and differenties it from cloud computing.

Ans:

Server virtualization is a techindogy that gives one Physical Server into multiple virtual machines each. virtual machine. Can run its own operation system. each machine. can Run its own operating System and application.

Role of virtualization:

- It improve hardware virtualization
- It reduce the need for physical Servers.
- It allows multiple os to run on one Server
- It helps allocate resource easily

Differences:

Virtualization	Cloud Computing
• It is a technology	• it is a Service model
• It create virtual maching	• It Provides Computing Services through a network
• Example: multiple vms on	• Example: Renting a Server

4. Explain the role of data centres in cloud computing

A data center is a facility containing services, storage device and networking equipment used to store and process data.

Roles of data Centers

1. Data Storage:
They store large amount of information
2. Data Processing:
They Processing user request and run applicatio
3. Networking:
They connect cloud server to user through the internet
4. Security:
They Protect data using security system.
5. Back up:
They maintain copies of important
6. Availability:
Back up system help reduce service interuption
7. Scalability:
They support addition resource when demand

5. Differentiate IaaS, PaaS, SaaS and Xaas with example.

Cloud Computing Provides different Services based on user requirement.

1. IaaS (Infrastructure as a Service)

- Users rent computing infrastructure and manage their operating System and applications.
- Provides Virtual Servers, Storage and networking.

Example: Amazon EC2.

2. PaaS (Platform as a Service)

- it Provides a Platform to develop and deploy application
- Developers need not manage the underlying infrastructure.

Example:

Google app engine

3. SaaS (Software as a Service)

- it Provides ready to use Software through the internet
- Users can access Software without installing and maintaining it themselves

Example:

4. XaaS (Everything as a Service)

- It Provids different IT resource and Service over the internet
- It include Services Such as storage database and Security

Example:

microSoft azore Services

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